MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 576A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Clipper diode or rectifier, high vacuum

See figure 1

Mounting position: Vertical, base down

Weight: 5-1/2-ounces (156 grams) nominal

ABSOLUTE RATINGS:

Parameter: Unit	Ef Vac	epx kv	ib a	Ieff mAdc (Note 1)	Ib mAdc	Pp W
Rectifier						
Maximum:	5.25	25.0	2.5	---	500	---
Minimum:	4.75	---	---	---	---	---
Clipper diode						
Maximum:	5.67	25.0	12.0	800	---	160
Minimum:	5.13	---	---	---	---	---
TEST CONDITIONS:	5.0	---	---	---	---	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	13.0	15.0	Aac
1231	Peak emission	-	eb = 5 kv; Ef = 5.4 Vac	0.65	II	is	14.0	---	a
1353	Operation of rectifiers (1)	2		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltages applied; t = 120 seconds	---	---	---	---	---	---
(D) 1041	Shock	-	Angle = 20°	---	---	---	---	---	---
1256	Electrode current (anode)	-	Eb = 200 Vdc	---	---	Ib	425	---	mAdc
1353	Operation of rectifiers (2)	3, 4	Ef = 5.4 Vac	---	---	ib	11.6	---	a
	Quality conformance inspection, part 3								
---	Life test (1)	-	Group C; operation of rectifiers (1); t = 500 hours	---	---	---	---	---	---
---	Life-test (1) end points: end points:								
1231	Peak emission	-		---	---	is	12.0	---	a
---	Failure to rectify	-		---	---	---	---	---	---
---	Life test (2)	3	Group D; operation of rectifiers (2); t = 50 hours	---	---	---	---	---	---
---	Life test (2) end points:								
1231	Peak emission	-		---	---	is	12.0	---	a
1353	Operation of rectifiers (2)	-		---	---	ib	11.0	---	a
1506	Heater-cycling life test	-	Group D; 1 minute "on", 1 minute "off"; Ef = 5.4 Vac	---	---	t	1,000	---	cycles
1231	Heater-cycling life-test end point:								
1231	Peak emission	-		---	---	is	14.0	---	a
1031	Variable-frequency vibration	5	No voltages applied	---	---	---	---	---	---

NOTES:

1. Effective anode current (Ieff) shall be measured with an rms indicator such as a thermocouple type meter.
2. Rectifier circuit: epx = 25 kv; ib = 2.5 a; Ib = 500 mAdc. The tube shall operate without signs of arcing or gas.
3. The tube shall be operated as a shunt diode in a thyratron modulator with the following circuit conditions: tp = 1.1 ±0.1 µs; network impedance = 53.0 ±1.0 ohms; thyratron mismatch inverse voltage (epx) = 23.9 kv minimum; peak thyratron forward voltage (epy) = 25 kv; prr = 525 ±5 percent; shunt diode series resistors = 1,750 ohms.
4. The tube shall operate for 2 minutes without signs of arcing or gas during a test interval not to exceed 5 minutes.

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NOTES: -Continued

(D) 5. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

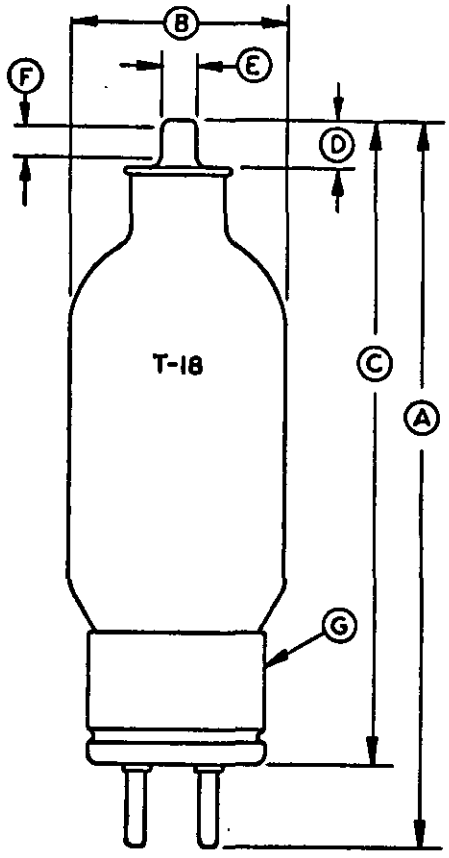
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2949)

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PIN CONNECTIONS	
Pin No.	Element
1	nc
2	f
3	f
4	nc
Cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	7.250 (184.15)	7.750 (196.85)
B	2.188 (55.58)	2.312 (58.72)
C	6.385 (162.18)	———
Quality conformance inspection, part 3 (see note 5)		
E	.370 (9.40)	.380 (9.65)
G	Base: A4-81 (EIA)	
Reference dimensions		
D	.500 (12.70)	
F	.313 (7.95)	

FIGURE 1. Outline drawing of electron tube type 576A.

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